Anatomy-Guided Masking Does Not Help: A Segmentation-Free Heuristic Outperforms a Trained Segmenter for Joint-Embedding Pretraining on Retinal OCT

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Abstract

Masking policy is an under-audited design choice in medical self-supervised learning. Intuitively, a masked-prediction model should be asked to reconstruct diagnostically relevant tissue, so masking guided by a trained segmentation model ought to beat unguided masking. We test that intuition directly. Starting from a single shared checkpoint, we continue patch-level I-JEPA pretraining on 3D retinal OCT under six masking policies that differ only in how predictor targets are placed, and evaluate all of them with one frozen linear probe protocol on glaucoma classification (FairVision, N=3000). The intuition fails. The best policy is an *oracle* band located by a first-order intensity statistic that consults no segmentation model (AUC 0.8855), beating the unguided null by $+0.0109$ (paired bootstrap 95% CI $[+0.0058, +0.0162]$, $p < 0.0005$) and beating a policy driven by a trained retinal segmenter by $+0.0047$. Policies that place targets most precisely on tissue — 98% of masked cells on anatomy — perform *worst*. We measure mask geometry for every arm and show that anatomical precision is not the operative variable, and we report a subgroup analysis in which Black participants have the lowest AUC in every valid probe. Encoder and probe weights are public and the headline number reproduces to 10^{-5} across hardware and numerical precision.

1 Introduction

Self-supervised pretraining is increasingly applied to medical imaging, where labels are scarce and anatomy is highly structured. I-JEPA [Assran et al., 2023] learns by predicting the representation of masked target blocks from a visible context block. Which patches become targets is a free design choice, and in the medical setting there is an obvious hypothesis: target the tissue that carries the diagnosis.

We test that hypothesis on glaucoma classification from 3D optical coherence tomography (OCT). All arms fork from one shared pretrained checkpoint at epoch 25 and differ *only* in masking policy thereafter, so any downstream difference is attributable to masking rather than to data, schedule, or initialisation.

Our contributions:

- A controlled six-arm comparison of masking policies for medical SSL, evaluated under one frozen-probe protocol (21 probes total).

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

Table 1: Frozen MeanPool probe, FairVision test (N=3000). All arms share one epoch-25 ancestor and one evaluation protocol. Δ is versus RANDOM at the same epoch.

arm	guidance signal	ep50	ep75	ep100
random	none (null)	0.8641	0.8723	0.8746
oracle	intensity centroid, no segmenter	0.8740	0.8836	**0.8855**
envelope	trained segmenter	0.8761	0.8803	0.8807
anatomy-v2	trained segmenter (shape)	0.8654	–	–
cover f=0.21	trained segmenter (coverage)	0.8643	–	–
Δ oracle − random		+0.0099	+0.0113	+0.0109
Δ oracle − envelope		−0.0020	+0.0033	+0.0047

- The finding that a segmentation-free intensity heuristic *beats* a trained-segmenter-guided policy, and that increasing anatomical precision of the mask *degrades* downstream AUC.
- Direct measurement of mask geometry for every arm, showing that "fraction of anatomy hidden" does not order the results.
- A subgroup audit, and a reproduction of the headline result to 10^{-5} from public weights on different hardware and precision.

2 Setup

Data. FairVision glaucoma OCT volumes. Downstream evaluation uses a fixed test split of N=3000 volumes (1466 positive / 1534 negative), seed 42, loader order fixed. Labels are byte-identical across arms, so any two arms are paired-comparable on the same cases.

Pretraining. Patch-level I-JEPA, ViT-Base, 256×256 crops, patch size 16 (a $16\times16=256$ token grid), predictor depth 6. Every arm resumes from the same epoch-25 checkpoint and continues to epoch 100 with identical optimiser, schedule, and effective batch size. Only the mask sampler differs.

Evaluation. The encoder is frozen. Slice features are mean-pooled, followed by LayerNorm and a linear head; the head is selected on validation AUC and reported on the held-out test split. This protocol is identical for all arms and all epochs.

2.1 Masking policies

random Stock I-JEPA multiblock: four uniform-random rectangles. No image content is consulted. This is the null.

oracle A band whose vertical position is located per slice by a per-column intensity-weighted row centroid, smoothed across columns. It uses a first-order intensity statistic only and consults **no segmentation model**.

envelope The same rectangles as RANDOM, but *placed* on a retinal envelope predicted by a trained segmentation model (MIRAGE). Shape, size, and count are unchanged; only location changes.

anatomy-v1 / anatomy-v2 Ragged connected components grown on the segmenter's score map, so target *shape* follows tissue. v2 adds diagonal bridging.

cover Targets chosen to cover the anatomy while leaving a fixed visible fraction (f=0.21) of tissue in the context.

3 Results

3.1 Anatomy-guided masking does not win

Table 1 gives frozen-probe test AUC at matched epochs.

Table 2: Measured mask geometry, 600 slices, production samplers. "purity" is the fraction of masked cells lying on tissue.

arm	anatomy hidden	purity	mask ratio	context kept	AUC
random	52.2%	31.6%	43.7%	42.1%	0.8746
oracle	62.2%	41.1%	40.0%	45.6%	**0.8855**
envelope	76.9%	43.5%	46.4%	40.5%	0.8807
cover f=0.21	74.1%	45.3%	43.3%	43.5%	0.8643
anatomy-v2	80.3%	**97.3%**	21.4%	67.9%	0.8654

The oracle band gives $+0.0109$ over the null at epoch 100 (paired bootstrap over test volumes, 95% CI $[+0.0058, +0.0162]$, $p < 0.0005$), and its margin is stable across epochs ($+0.0099$, $+0.0113$, $+0.0109$). The segmenter-guided ENVELOPE arm also beats the null, but by a margin that *shrinks* with training ($+0.0120 \rightarrow +0.0080 \rightarrow +0.0062$), and it is overtaken by the segmentation-free oracle by epoch 75.

3.2 Anatomical precision is not the operative variable

We measured mask geometry directly, running each production sampler on 600 held-out slices (Table 2).

The ordering is not explained by anatomical targeting. The anatomy arms place 97–98$\%$ of masked cells on tissue and rank last; the oracle places 41% on tissue and ranks first. Across the four rectangle arms, Spearman correlation between fraction-of-anatomy-hidden and AUC is -0.40 — the wrong sign — and no measured geometric variable is a significant predictor at this sample size.

Two mechanisms are consistent with the data and are not separated by these experiments. First, the oracle is the most *spatially consistent* policy we measured (highest concentration of its hit map, Gini 0.400 versus 0.316–0.338 for the other rectangle arms): it repeatedly forces the encoder to represent the same region. Second, the anatomy arms differ from the rectangle arms on several axes at once — they leave 68% of tokens as context versus 42%, and they run at roughly half the effective masking ratio — so their deficit cannot be attributed to target *shape* alone. We therefore do not claim that irregular target shape is harmful; that comparison is confounded by construction and we report it as such.

3.3 Subgroup analysis

Joining saved test predictions to demographics in deterministic loader order (validated by exact label reconstruction), Black participants have the lowest AUC in every valid probe. The disparity is a property of the evaluation, not of a single arm: it persists across masking policies and across epochs, which suggests it is inherited from the data and the downstream task rather than introduced by the pretraining objective.

3.4 Reproducibility

The epoch-100 oracle encoder and its trained probe head are public. Re-encoding the test split on different hardware, at fp32 rather than fp16, and with a different encoder chunk size reproduces the reported AUC to 9.8×10^{-6} (0.8854754 versus 0.8854852), a difference of 22 discordant pairs out of 2,248,844. Probe-time numerical precision is therefore not a material factor in any comparison reported here.

4 Limitations

One pretraining run per policy. The paired bootstrap quantifies sampling error on a fixed test set; it does not quantify seed-to-seed variance of pretraining. With n=1 continuation per arm, the *ranking* of arms is not established even where individual pairwise differences are significant on this test set. This is the principal limitation of the study.

One visible-tissue floor. Only $f=0.21$ was pretrained for the COVER policy, so we report no dose-response over the floor and make no claim that any particular floor is optimal.

Confounded shape comparison. As noted above, the anatomy arms differ from the rectangle arms in masking ratio, context size, and the number of predictor loss terms simultaneously. We report their results but do not use them to support a causal claim about target shape.

Single dataset and task. All results are glaucoma classification on FairVision OCT. Generality to other modalities or endpoints is untested.

5 Conclusion

Guiding masked-prediction targets with a trained segmentation model did not improve downstream glaucoma classification relative to a far cheaper segmentation-free heuristic, and increasing the anatomical precision of the mask was associated with worse, not better, representations. For practitioners the practical implication is concrete: an entire segmentation stage can be removed from this pipeline without measured loss. For the field, the result is a caution that "mask the clinically relevant region" is an intuition that should be tested rather than assumed.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15619–15629, 2023. doi: 10.1109/CVPR52729.2023.01499.